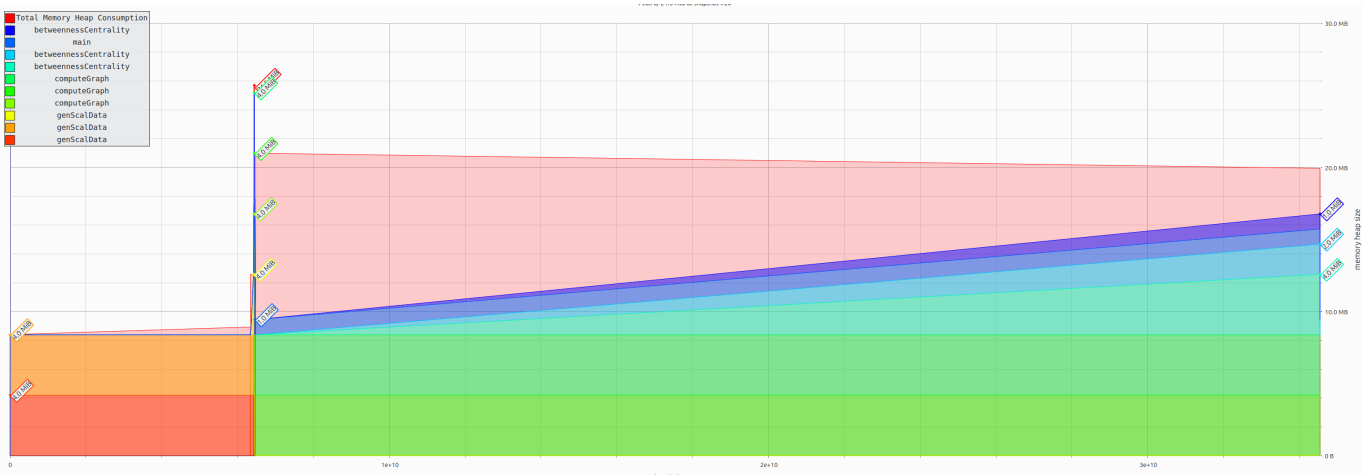
Exercise 04 - Robert Zacchia

A) Memory profiling

- Use the valgrind "massif" tool in Valgrind to determine the largest sources of heap memory utilization, and visualize the results with "massif-visualizer".



- How significant is the perturbation in execution time caused by using massif?

As seen in the table below, profiling the memory of a program has a significant overhead. About two times the RAM is allocated just for the profiling, and the execution time is also about double.

Massif	local (wtime, peak RAM)	lcc3 (wtime, peak RAM)
without massif	23.84 s, 26992 kB	32.29 s, 27364 kB
with massif	40.89 s, 54264 kB	68.58 s, 56348 kB

B) Measuring CPU counters

For both programs, measure **all** events in the `[Hardware cache event]` category reported by `perf list`. Note that as discussed in the lecture, there is a limit on the number of hardware counters you can measure in a single run.

cpu information

Model name: Intel(R) Xeon(R) CPU X5650 @ 2.67GHz CPU MHz: 2793.000 L1d cache: 32K L1i cache: 32K L2 cache: 256K L3 cache: 12288K

Overview of Hardware cache event counters

L1 Data Cache (L1-dcache)

- L1-dcache-loads: Number of loads (reads) from main memory into the L1 data cache
- L1-dcache-load-misses: Data cache misses on load operations (data not in L1, must fetch from higher cache levels)
- L1-dcache-stores: Number of stores (writes) to the L1 data cache

- L1-dcache-store-misses: Data cache misses on store operations
- L1-dcache-prefetches: Hardware prefetch requests for the L1 data cache (automatic data fetching ahead of time)
- L1-dcache-prefetch-misses: Prefetch requests that failed to find data in L1
- L1 Instruction Cache (L1-icache)
- L1-icache-loads: Number of instruction fetches from the L1 instruction cache
- L1-icache-load-misses: Instruction cache misses (instruction not in L1, must fetch from higher cache levels)

Last-Level Cache (LLC) — typically L3 cache

- LLC-loads: Loads from the last-level cache (when L1 misses)
- LLC-load-misses: Misses in the last-level cache (must go to main memory)
- LLC-stores: Stores to the last-level cache
- LLC-store-misses: Store misses in the last-level cache
- LLC-prefetches: Hardware prefetch requests for the LLC
- LLC-prefetch-misses: LLC prefetch requests that failed

Data TLB (dTLB) — Translation Lookaside Buffer for data access

- dTLB-loads: Virtual-to-physical address translations for loads
- dTLB-load-misses: Address translation misses on loads (must walk page tables)
- dTLB-stores: Virtual-to-physical address translations for stores
- dTLB-store-misses: Address translation misses on stores

Instruction TLB (iTLB)

- iTLB-loads: Virtual-to-physical address translations for instruction fetches
- iTLB-load-misses: Address translation misses for instructions

NUMA Node Counters — for multi-socket systems

- node-loads: Memory loads from the local NUMA node
- node-load-misses: Loads that had to access remote NUMA nodes
- node-stores: Memory stores to the local NUMA node
- node-store-misses: Stores to remote NUMA nodes
- node-prefetches: Prefetch requests on local node
- node-prefetch-misses: Prefetches that missed locally

Branch Prediction

- branch-loads: Total branch instructions executed
- branch-load-misses: Branch prediction misses (incorrect predictions requiring pipeline flushes)

For both programs:

- Report the results in **relative** metrics, and compare these between the programs.

Performance Comparison: SSCA2 vs NPB-BT-A

Metric	SSCA2	NPB-BT-A
L1-dcache-load-misses	38.54% (±0.02%)	3.98% (±0.11%)
L1-dcache-store-misses	24.75% (±0.00%)	3.20% (±0.04%)
L1-dcache-prefetch-misses	74.07% (±0.08%)	37.95% (±0.24%)
L1-icache-load-misses	0.00% (±7.41%)	0.04% (±1.60%)
LLC-load-misses	10.38% (±0.51%)	49.23% (±0.32%)
LLC-store-misses	2.11% (±0.94%)	6.24% (±3.28%)
LLC-prefetch-misses	21.45% (±3.64%)	62.39% (±0.73%)
dTLB-load-misses	13.50% (±0.23%)	0.00% (±0.78%)
dTLB-store-misses	10.85% (±0.24%)	0.0005% (±0.38%)
iTLB-load-misses	0.00% (±8.01%)	0.00% (±8.95%)
node-load-misses	0.0006% (±40.91%)	0.0015% (±14.33%)
node-store-misses	0.00% (±0%)	0.00% (±0%)
node-prefetch-misses	0.008% (±87.61%)	0.0004% (±337.91%)
branch-load-misses	200.40% (±0.29%)	74.33% (±0.33%)

- How significant is the perturbation in execution time caused by using perf to measure performance counters?

Run / Counter group	SSCA2	NPB-BT-A
L1-dcache	`32.1512 ±0.0514 s`	`76.023 ±0.146 s`
L1-icache	`32.317 ±0.179 s`	`76.382 ±0.502 s`
LLC	`32.213 ±0.252 s`	`76.476 ±0.135 s`
LLC-prefetch	`32.093 ±0.313 s`	`76.002 ±0.118 s`
dTLB	`32.2115 ±0.0336 s`	`76.037 ±0.178 s`
iTLB	`29.18 ±1.85 s`	`76.947 ±0.724 s`
node	`32.4932 ±0.0339 s`	`76.597 ±0.132 s`
node-prefetch	`30.25 ±2.03 s`	`75.965 ±0.106 s`
branch	`31.431 ±0.734 s`	`76.035 ±0.234 s`
baseline (no counters)	`31.673 ±0.862 s`	`76.766 ±0.122 s`